Python Quick Start Guide for Windows: Mech. Sensors Trainer

STEP 1 Check Components and Details

Make sure your Mechatronic Sensors Trainer case includes the following components.

1. Mechatronic Sensors Trainer
2. Base
3. USB A – C and C – C Cable
4. Force Sensing Resistor
5. Thermocouple
6. Potentiometer
7. Radar Cone
8. Load Cell and Scale Pieces
9. Sensors Trainer Resources. Content and courseware provided digitally at github.com/quanser/Quanser_Academic_Resources

Ensure you local computer has the following components

1. Windows 10/11 operating system
2. Quanser SDK 2026 or QUARC 2026 or later for either, installed

STEP 2 Download Content and Set Up Computer

A Download the resources for the device from https://github.com/quanser/Quanser_Academic_Resources
Follow the instructions on the 'Downloading Resources' and 'Setting Up Your Computer' sections.

B You should now have a supported Python version downloaded and added to PATH and the 'Completing the Setup' section from 'Setting Up Your Computer' should have required a restart of your PC after installing all necessary libraries.

C Download VS Code or a code editor of your choice. If you use VS Code, please install the Python extension (VS code will suggest it once you open the python file in step 4). You should now have all the necessary files to run the device.

STEP 3 Set Up the Hardware

The steps below outline the instructions to setup the Mechatronic Sensors Trainer.

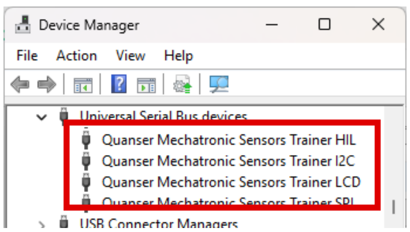
A

Connect one of the provided **USB C connector** to the device and then to an enabled USB 3.0 (or higher) port on your computer.

The power LED next to the connector should turn to white and the LCD screen should load the Quanser Logo and then show a black background with white crosses in a grid. LCD will say Low-power mode if USB A to C was used. That is expected, see the user manual for more information.

If step 2 was done successfully and QUARC or Quanser SDK were properly installed, Windows[1] should automatically detect the presence of the Sensors Trainer.

B



Wait for around a minute and go to Windows Device Manager and verify that the device appears as multiple devices (HIL, I2C, LCD, …) under *Universal Serial Bus devices*. **If they appear as *Other devices* instead with a question mark next to them**, right click on them one at a time and click *Uninstall device* and confirm the pop up. Once they all disappear from that list, unplug your device for 30 seconds and plug it in again. The device should now properly appear as USB devices.

STEP 4 Testing the Mechatronic Sensors Trainer

Follow the procedure below to test your Sensors Trainer.

A

Open `quick_start_sensors.py` on your code editor (VS Code) and click run.

B

1. This code is meant to print the encoder counts in the terminal every 0.5 seconds.
2. If it fails while initializing and loading any library, it will print instructions on how to set up the system.
3. If it starts printing, rotate the encoder knob next to the LCD screen and observe the counts change. Your device should be ready to use!
4. The code automatically stops after 200 seconds, or you can stop it using Ctrl+C on the terminal.

```
/2_quick_start_guides/mech_sensors_trainer/quick_start_sensors.py"
Encoder counts: 0
Encoder counts: 1
Encoder counts: 3
Encoder counts: 6
Encoder counts: 6
Encoder counts: 4
Encoder counts: 1
Encoder counts: -2
Encoder counts: -5
Encoder counts: -7
Encoder counts: -7
KeyboardInterrupt handled. Cleaning up.
Sensors Trainer closed gracefully.
```

TROUBLESHOOTING

Review the following recommendations before contacting Quanser's technical support engineers.

1. Check the connections outlined in Step 3 of this guide and ensure that the cables are connected firmly.
2. If the device is turned on properly and an error happens when running the python code, follow the instructions printed on the terminal to understand what could have gone wrong.

LEARN MORE To browse and download the latest Quanser resources visit https://github.com/quanser/Quanser_Academic_Resources

STILL NEED HELP For further assistance from a Quanser engineer, contact us at tech@quanser.com

[1] Windows is a registered trademark of Microsoft Corporation in the United States and other countries.